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System Monitor Tool

Objective: Create a system monitor tool in C++ that displays real-time information about system processes, memory usage, and CPU load, similar to the 'top' command.

Day-wise Tasks:

Day 1: Design UI layout and gather system data using system calls.

```
swayam@DESKTOP-744JU4C:~/Desktop/2141016281_lsp$ g++ day01.cpp -o day01
swayam@DESKTOP-744JU4C:~/Desktop/2141016281_lsp$ ./day01
Total RAM: 7897 MB
Total RAM: 7 GB
Free RAM : 6959 MB
Free RAM : 6 GB
swayam@DESKTOP-744JU4C:~/Desktop/2141016281_lsp$
```

Day 2: Display process list with CPU and memory usage.

```
swayam@DESKTOP-744JU4C:~/Desktop/2141016281_lsp$ g++ day02.cpp -o day02
swayam@DESKTOP-744JU4C:~/Desktop/2141016281_lsp$ ./day02
CPU Usage: 0%

Raw CPU Values:
user: 439
nice: 0
system: 532
idle: 378440
iowait: 600
irq: 0
softirq: 83
steal: 0
guest: 0
guest_nice: 0
Total CPU Time: 380094
Idle Time: 379040
Manual CPU Usage Calculation: 0.2773%
swayam@DESKTOP-744JU4C:~/Desktop/2141016281_lsp$
```

Day 3: Implement process sorting by CPU and memory usage.

```
swayam@DESKTOP-744JU4C:~/Desktop/2141016281_lsp$ g++ day03.cpp -o day03
swayam@DESKTOP-744JU4C:~/Desktop/2141016281_lsp$ ./day03
PID      CPU%      Memory (kb)      Name
9        0.669539%      4607      (init)
1        0.0669539%      4607      (systemd)
52       0.0276481%      4607      (systemd-journal)
75       0.0215041%      4607      (systemd-udev)
143      0.018432%      11740     (systemd-resolve)
591      0.017802%      8024      (polkitd)
203      0.0154074%      6492      (unattended-upgr)
161      0.01536%       6492      (wsl-pro-service)
334      0.0123259%      6492      (systemd)
158      0.012288%      6492      (systemd-logind)
swayam@DESKTOP-744JU4C:~/Desktop/2141016281_lsp$
```

Day 4: Add functionality to kill processes.

```
swayam@DESKTOP-744JU4C:~/Desktop/2141016281_lsp$ g++ day04.cpp -o day04
swayam@DESKTOP-744JU4C:~/Desktop/2141016281_lsp$ ./day04
PID      CPU%      Memory (kb)      Name
9        0.553483%      4607      (init)
1        0.0553483%      4607      (systemd)
52       0.0226387%      4607      (systemd-journal)
75       0.0176078%      4607      (systemd-udev)
143      0.0150924%      11740     (systemd-resolve)
203      0.0126088%      6492      (unattended-upgr)
161      0.012577%       6492      (wsl-pro-service)
591      0.0124714%      8024      (polkitd)
334      0.010087%       6492      (systemd)
158      0.0100616%      6492      (systemd-logind)
Enter PID to kill :75
Failed to kill process: Operation not permitted
swayam@DESKTOP-744JU4C:~/Desktop/2141016281_lsp$
```

Day 5: : Implement real-time update feature to refresh data every few seconds.

```
PID      CPU%      Memory (kB)      Name
9        0.396099      0      (init)
1        0.0382785      0      (systemd)
52       0.0188725      0      (systemd-journal)
75       0.0120097      0      (systemd-udev)
143      0.0102941      11740     (systemd-resolve)
203      0.00859313      0      (unattended-upgr)
161      0.00857839      0      (wsl-pro-service)
591      0.00704457      8024      (polkitd)
334      0.00687451      0      (systemd)
282      0.00687451      0      (bash)
Enter PID to Kill :
591
Failed to kill Process: Operation not permitted

Press 'q' to quit or Enter to refresh :
```